

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 7 Molecular Biology of Microbial Growth

7.1 Multiple Choice Questions

1) Fluorescence microscopy allows visualization of all of the following EXCEPT

- A) cytoskeletal proteins.
- B) nuclear proteins.
- C) interactions between cytoplasmic proteins.
- D) All of these answer choices can be visualized.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.1

2) Once replication of the chromosome has begun, DnaA is inactivated

- A) as a result of being blocked from *oriC* binding.
- B) by increased expression of FtsZ.
- C) as a result of increased binding to ATP.
- D) by increased binding of DnaA-ATP to *oriC*.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.2

3) _____ prohibits re-initiation of chromosome replication.

- A) DnaA-ATP
- B) SeqA
- C) *oriC*
- D) FtsZ

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.2

4) Prior to DNA replication, both strands of the chromosome are methylated on the _____ residue of the sequence-GATC-.

- A) A
- B) G
- C) C
- D) T

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.2

- 5) Hemimethylated DNA is found on
- A) the newly synthesized strand of DNA immediately after replication.
 - B) the parental strand of DNA immediately after replication.
 - C) both strands of dsDNA immediately after replication.
 - D) neither strand of DNA immediately after replication.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.2

- 6) The typical time required for chromosome replication in *Escherichia coli* is
- A) 10 minutes.
 - B) 20 minutes.
 - C) 30 minutes.
 - D) 40 minutes.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.2

- 7) The typical generation time of *Escherichia coli* is approximately _____ under optimal environmental conditions.
- A) 10 minutes
 - B) 20 minutes
 - C) 30 minutes
 - D) 40 minutes

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.2

- 8) The Par system is necessary for
- A) formation of a replication fork in *Escherichia coli*.
 - B) distribution of genetic material in replicating *Caulobacter*.
 - C) septum formation.
 - D) elongation of bacillus prior to cell division.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.2

- 9) Which of the following proteins is most active in divisome complexes?
- A) FtsZ
 - B) crescentin
 - C) MinCD
 - D) MreB

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.3

10) A bacterium that lacks the *mreB* gene will have a _____ shape.

- A) bacillus
- B) coccoid
- C) short bacillus
- D) vibrio

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.4

11) What are the primary regulator units that control endospore formation?

- A) allosteric proteins
- B) antisense RNAs
- C) riboswitches
- D) sigma factors

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.6

12) All of the following are TRUE statements concerning binary fission of microbial cells EXCEPT

- A) the chromosome of the cell is replicated.
- B) elongation of the cell occurs and the chromosomes are pushed apart.
- C) a septum is formed across the midline of the cell.
- D) daughter cells produced can be of different sizes.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.6

13) There are four basic stages to biofilm formation. Which of the following is the correct order of these stages?

- A) attachment, colonization, development, dispersal
- B) colonization, attachment, development, dispersal
- C) development, colonization, attachment, dispersal
- D) attachment, development, colonization, dispersal

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.9

14) *Pseudomonas aeruginosa* is a(n) _____ pathogen.

- A) opportunistic
- B) idiopathic
- C) pessimistic
- D) iatrogenic

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.9

15) Which of the following activities is a result of cyclic di-guanosine monophosphate synthesis?
A) transition from planktonic to sessile growth during biofilm formation
B) reduction in activity of flagellar motor
C) biosynthesis of extracellular matrix during biofilm formation
D) All of these answer choices activities are a result of cyclic di-guanosine monophosphate synthesis.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.9

16) The major symptoms of the genetic disorder cystic fibrosis are caused due to biofilm formation by

- A) *Serratia marcesans*.
- B) *Pseudomonas aeruginosa*.
- C) *Vibrio cholera*.
- D) *Bacillus subtilis*.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.9

17) All of the following are targets of currently used antibiotics EXCEPT

- A) DNA synthesis.
- B) biofilm formation.
- C) transcription.
- D) translation.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.10

18) When a *Bacillus anthracis* population suddenly must form spores to survive a harsh nutrient poor environment, how do the cells obtain energy?

- A) Cells in a growth phase that have not used up all of their energy will be the only ones to make endospores, which is why relatively few endospores are often made from a large population.
- B) Intracellular energy reserves are quickly made available to produce endospores.
- C) Slow responding cells are cannibalized by others that already began spore formation.
- D) Global regulation is initiated to minimize energy waste in biosynthetic pathways and catabolic pathways are increased to consume remaining usable substrates to fuel spore formation.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.6

19) Bacteria from the genus *Caulobacter* are used to model cellular differentiation in eukaryotes. The abundance of CtrA, DnaA, and GcrA separately control activity of other genes necessary for differentiation in *Caulobacter*. Thus, these three proteins can be classified as

- A) activating sensors.
- B) heterologous regulators.
- C) differentiating regulons.
- D) transcriptional regulators.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.7

20) A bacterium that either partially or fully catabolizes an acyl-homoserine lactone will likely disrupt

- A) attenuation.
- B) chemotaxis.
- C) endospore formation.
- D) quorum sensing.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.9

21) Quorum sensing generally follows the mechanism of which type of regulation?

- A) feedback inhibition
- B) negative transcriptional regulation
- C) positive transcriptional regulation
- D) a two-component regulation system

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.9

22) Penicillins work by

- A) activating autolytic enzymes.
- B) blocking the transpeptidation reaction that forms peptidoglycan cross-links.
- C) stimulating autolysins to form holes in the plasma membrane.
- D) All of these answer choices are true.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.10

23) Differential gene expression occurs in each of the following bacterium EXCEPT

- A) *Bacillus*.
- B) *Mycobacterium*.
- C) *Anabaena*.
- D) *Caulobacter*.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.6

24) σ^F functions as a(n) _____ during endospore formation.

- A) transcription factor
- B) anti-sigma factor
- C) protease
- D) RNase

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.6

25) When Spo0A is highly phosphorylated

- A) heterocyst formation is initiated.
- B) stalk formation is initiated.
- C) endospore formation is initiated.
- D) photosynthesis is initiated.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.6

26) What is the function of heterocysts in cyanobacteria such as *Anabaena*?

- A) nitrogen fixation
- B) sexual reproduction
- C) asexual reproduction
- D) photosynthesis

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.6

27) The major shape-determining factor in *Bacteria* is the protein

- A) FtsZ.
- B) crescentin.
- C) MreB.
- D) MinD.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.3

28) In the process of binary fission in *Bacteria*, which action occurs first?

- A) formation of the divisome
- B) DNA replication
- C) cell elongation
- D) cytokinesis

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.3

29) FtsZ is related to _____, an important protein involved in cell division.

- A) actin
- B) myosin
- C) tubulin
- D) collagen

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.3

30) FtsA is related to _____, an important cytoskeletal protein in eukaryotes.

- A) actin
- B) myosin
- C) tubulin
- D) collagen

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.3

31) Endospore formation is stimulated when

- A) bacterial growth ceases due to limitation of an essential nutrient.
- B) the bacterium is undergoing binary fission.
- C) bacteria are dividing exponentially.
- D) nutrient levels rise.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.6

32) Proteins MinC, MinD, and MinE interact to

- A) help guide FtsZ to the cell midpoint.
- B) initiate peptidoglycan synthesis.
- C) assist chromosome segregation.
- D) determine cell shape.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.4

33) The curved-rod shape typical of *Caulobacter* is due to the protein(s)

- A) crescentin.
- B) MreB.
- C) crescentin and MreB.
- D) neither crescentin nor MreB.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.4

34) Which of the following traits is NOT characteristic of the *Caulobacter* life cycle?

- A) Cell division only occurs in the swarmer stage.
- B) Cell division only occurs in the stalked cell stage.
- C) Chromosome replication only occurs in the swarmer stage.
- D) Chromosome replication and cell division only occurs in the stalked cell stage.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.7

35) Biofilm formation in *Pseudomonas aeruginosa* is triggered by _____ cell densities and repressed by _____ cell densities.

- A) high; high
- B) low; low
- C) high; low
- D) low; high

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.9

36) Biofilm formation in *Vibrio cholera* is triggered by _____ cell densities and repressed by _____ cell densities.

- A) high; high
- B) low; low
- C) high; low
- D) low; high

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.7

37) What is the function of bactoprenol?

- A) It is a hydrophobic alcohol that transports peptidoglycan precursors across the cytoplasmic membrane.
- B) It is responsible for forming the peptide cross-links between muramic acid residues in adjacent glycan chains.
- C) It triggers the recruitment of FtsZ and the initiation of the divisome.
- D) It supplies the energy necessary for transpeptidation to occur.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.5

38) Shortage of key nutrients such as _____ results in transcription and secretion of a toxic protein that leads to cannibalization of neighboring cells of the same species.

- A) magnesium
- B) phosphate
- C) calcium
- D) iron

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.6

39) Heterocysts are specialized cells that undergo

- A) oxidative phosphorylation.
- B) photophosphorylation.
- C) nitrogen fixation.
- D) lysine decarboxylation.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.8

40) Which of the following is NOT a function of DnaA in *Caulobacter*?

- A) initiation of DNA replication
- B) initiation of swarming
- C) transcriptional regulation
- D) All of these answer choices are functions of DnaA.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.7

41) Why is the presence of a cell wall significant from a clinical standpoint?

- A) All types of cells have a cell wall, and it makes identification of the causative agent of disease difficult.
- B) The cell wall protects microorganisms from destruction by the immune system.
- C) Animal cells do not have cell walls, so antibiotics that target cell walls can selectively destroy invading microorganisms.
- D) Only gram-negative *Bacteria* have cell walls.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.5

42) Which of the following molecules functions as an intracellular signaling molecule during quorum sensing?

- A) acylated homoserine lactones
- B) hydrophilic lipids
- C) quinones
- D) proteorhodopsins

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.9

43) Dormancy is a result of all of the following EXCEPT

- A) phenotypic heterogeneity.
- B) ineffective efflux pumps.
- C) TA modules.
- D) the stringent response.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.11

7.2 True/False Questions

1) The purpose of hemimethylated DNA is to block chromosome replication.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.2

2) Plasmids are replicated in *Escherichia coli* using the Par system.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.2

3) PopZ plays a critical role in localizing proteins to the cell poles.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.2

4) The Par system is analogous to spindle fibers in eukaryotic cells.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.2

5) In the *Caulobacter* life cycle, the role of swarmer cells is strictly for reproduction.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.7

6) Biofilm formation is more likely to occur when *Vibrio cholerae* is found in low densities such as sea water.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.9

7) Multiple sigma factors are essential to induce the biosynthesis of endospores, consequently a complex regulatory mechanism such as this has a higher chance of mutation leading to incorrect functioning compared to a simple repression mechanism.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.6

8) Generation of new bacillus-shaped cells starts with one cell elongating and terminates when it splits into two separate bacilli.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.4

9) The activity of MinC and MinD direct whether a bacterial cell will be coccoid or bacillus shaped.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.4

10) In fast-growing *Escherichia coli* cells, multiple replication forks of genomic DNA allows binary fission to occur before the genome has been fully duplicated.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.2

11) Multiple sigma factors are essential to induce the formation of endospores.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.6

12) During elongation of a cell during binary fission, small gaps caused by transglycosylases are formed before cell membrane precursors can be inserted.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.3

13) Antibiotic resistance can develop in a bacterium as a result of spontaneous mutation.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.10

14) Natural antibiotics are produced by all bacterial genera.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.10

15) All members of the genus *Pseudomonas* respond similarly to c-di-GMP.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.9

16) The spherical shape of *Staphylococcus aureus* is a result of MreB function.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.4

17) Both crescentin and MreB play a role in formation of the vibrio-shaped bacterium *Caulobacter*.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.4

18) FtsZ is analogous to the protein tubulin in eukaryotic cells.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.4

19) In both cocci as well as rod-shaped cells, new peptidoglycan synthesis occurs in both directions from the central location of the FtsZ ring.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.5

20) Oxygenic phototrophs require the uptake of oxygen for the reactions of photosynthesis.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.8

21) Mobile resistance genes encode enzymes that inactivate an antibiotic by altering its structure.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.10

22) Unlike penicillin, methicillin targets an alternative penicillin binding protein called MecA in nonresistant *Staphylococcus aureus*.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.10

23) MRSA strains synthesize MecA only in the presence of methicillin or other β -lactam antibiotics.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.10

24) Antibiotic-sensitive bacteria sometimes produce cells that are transiently resistant to multiple antibiotics.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.10

25) When *Mycobacterium tuberculosis* is in the dormant state it is still susceptible to antibiotics, but the rate of death is slower.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.11

26) Persistence is a heritable trait.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.11

27) The toxin component of TA modules helps ensure survival during stressful growth conditions.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.11

28) Free HipA toxin inhibits transcription.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.11

7.3 Essay Questions

1) Describe the functions of the FtsZ protein and the Z-ring in bacterial cells.

Answer: FtsZ is an essential cell division protein that forms a contractile ring structure (Z ring) at the future cell division site. The regulation of the ring assembly controls the timing and the location of cell division. One of the functions of the FtsZ ring is to recruit other cell division proteins to the septum to produce a new cell wall between the dividing cells.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.3

2) What is the purpose of the MreB protein in bacteria?

Answer: The bacterial actin homologue MreB is required for the maintenance of a rod-shaped cell and has been shown to form spirals that traverse along the longitudinal axis of *Bacillus subtilis* and *Escherichia coli* cells.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.4

3) What role does the Par system play in bacterial cell division?

Answer: The Par (partitioning) system can be found in many bacteria including *Caulobacter* which undergoes budding. Similar to mitotic spindle fibers which separate replicated chromosomes in eukaryotic cells, the Par system distributes chromosomes and plasmids equally to progeny cells during growth. This system is composed of the ParAATPase, the ParB chromosome-binding protein, and the PopZ complex, as well as a centromere-like parS sequence located near *oriC*.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.2

4) A researcher unexpectedly identified mutant bacteria deficient in the FtsZ protein. Their growth appeared filamentous, and displayed incomplete cell division. Explain the role of the FtsZ protein and how a deficiency would account for this altered growth.

Answer: FtsZ is an essential cell division protein that forms a contractile ring structure (Z ring) at the future cell division site. The regulation of the ring assembly controls the timing and the location of cell division. One of the functions of the FtsZ ring is to recruit other cell division proteins to the septum to produce a new cell wall between the dividing cells. Consequently, a mutation in this protein would result in incomplete cell division and a potentially filamentous appearance.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.3

5) How might a genetically modified strain of *E.coli* with a nonfunctional *mreB* gene compare to *E. coli* possessing a functional *mreB* gene?

Answer: MreB forms a simple cytoskeleton in *Bacteria* by forming patchlike filaments around the inside of the cell just below the cytoplasmic membrane. The MreB cytoskeleton is thought to define cell shape by recruiting other proteins that function in cell wall growth to group in a specific pattern. Inactivation of the gene encoding MreB in rod-shaped bacteria causes the cells to become coccus-shaped.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 7.3

6) Explain how FtsZ-type proteins being found in mitochondria and chloroplasts strengthens or weakens the endosymbiosis theory.

Answer: The protein FtsZ, encoded by the gene *ftsZ*, is specifically involved in prokaryotic cell division. FtsZ-like proteins identified in mitochondria and chloroplasts suggest the FtsZ-encoding gene was first present in prokaryotes and that these two organelles were subsequently developed and maintained in Eukarya. Ultimately this corroborates the ideas of the endosymbiotic theory and seemingly links chloroplast and mitochondrial origins to prokaryotes.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 7.3

7) Explain the role of CtrA in the formation of swarmer cells in *Caulobacter*.

Answer: CtrA is activated by phosphorylation in emerging swarmer cells in response to external signals. Once phosphorylated, CtrA-P activates genes that encode synthesis of the flagellum and other functions specific to swarmer cells. Conversely, CtrA-P represses the synthesis of the protein CcrA and so inhibits the initiation of DNA replication in swarmer cells by binding to and blocking the origin of replication.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.7

8) Explain how green fluorescent protein (GFP) could be utilized to determine the role of FtsZ in binary fission in *Escherichia coli*.

Answer: For visualizing molecular events, reporter genes that encode fluorescent products such as green fluorescent protein (GFP) are routinely used. In this case, the gene for the production of GFP would be incorporated into the genome of *Escherichia coli* in the region of the DNA that codes for the FtsZ gene so that both genes are controlled by the same regulatory sequence; that is, the gene's regulatory sequence now controls the production of GFP, in addition to FtsZ. In cells where the gene is expressed, and the tagged FtsZ proteins are produced, GFP is produced at the same time. Thus, only those cells in which the tagged gene is expressed, or the target proteins are produced, will fluoresce when observed under fluorescence microscopy.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 7.3

9) Explain the impact spontaneous mutation of the gene *minE* in an *Escherichia coli* bacterium would have on future cell divisions.

Answer: The MinE protein is one of three proteins of the Min system encoded by the *minB* operon required to generate pole to pole oscillations prior to bacterial cell division as a means of specifying the midzone of the cell. If a spontaneous mutation were to occur resulting in inactivation of this protein, it would affect cell division in the next generation since the FtsZ ring would not form and cell division would not occur.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 7.3

10) Compare the process of new peptidoglycan formation in cocci versus rod-shaped cells.

Answer: In cocci, new cell wall material grows out in opposite directions from the FtsZ ring. In contrast, new cell wall grows at several locations along the length of a rod-shaped cell.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.5

11) The chemical composition of the peptidoglycan carrier bactoprenol is highly hydrophobic. Explain why this is necessary for bactoprenol to carry on its function.

Answer: Bactoprenol is a hydrophobic C55 alcohol that bonds to peptidoglycan precursors.

Bactoprenol transports peptidoglycan precursors across the cytoplasmic membrane by rendering them sufficiently hydrophobic to pass through the fatty acid tails of the membrane interior. Once in the periplasm, bactoprenol interacts with enzymes called transglycosylases that insert cell wall precursors into the growing point of the cell wall and catalyze glycosidic bond formation.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.5

12) How does the antibiotic penicillin lead to the death of pathogens?

Answer: Several penicillin binding proteins have been identified in bacteria, including FtsI.

When penicillin is bound to penicillin-binding proteins, the proteins lose their catalytic activity. In the absence of transpeptidation, the continued activity of autolysins so weakens the cell wall that the cell eventually bursts.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.5

13) Describe the steps of binary fission.

Answer: 1) Replication of the circular prokaryotic chromosome begins at the origin of replication and continues in both directions at once. 2) The cell begins to elongate. FtsZ proteins migrate toward the midpoint of the cell. 3) The duplicated chromosomes separate and continue to move away from each other toward opposite ends of the cell. FtsZ proteins form a ring around the periphery of the midpoint between the chromosomes. 4) The FtsZ ring directs the formation of a septum that divides the cell. Plasma membrane and cell wall materials accumulate. 5) After the septum is complete, the cell pinches in two, forming two daughter cells. FtsZ is dispersed throughout the cytoplasm of the new cells.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 7.3

14) Explain why the antibiotic penicillin is effective at destroying bacteria but does not have a similar effect in eukaryotic cells.

Answer: Animal cells do not possess a cell wall and therefore lack peptidoglycan. Consequently the drug can be administered in high doses and is typically nontoxic. Most pathogenic bacteria do have cell walls composed of peptidoglycan and are thus potential targets of the drug.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.5

15) The major trigger of endospore formation in *Bacillus* is nutrient deprivation. Without the appropriate nutrients, how can endospore formation occur?

Answer: Sporulating cells cannibalize cells of their own species to obtain the appropriate nutrients. Those *Bacilli* in which Spo0A has become activated secrete a toxic protein that lyses nearby *Bacillus* cells whose Spo0A has not yet become activated. The lytic protein is produced in addition to another protein that functions to delay sporulation of neighboring cells. Cells committed to sporulation also make an antitoxin protein to protect themselves against the effects of their own toxic protein. Once lysed, the sacrificed cells are used as a source of nutrients.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.6

16) Nitrogen fixation utilizes the enzyme nitrogenase, which is extremely sensitive to oxygen. How do cyanobacteria carry on both oxygenic photosynthesis and nitrogen fixation at the same time?

Answer: Some filamentous cyanobacteria such as *Anabaena* and *Nostoc* form specialized cells called heterocysts that are exclusively dedicated to nitrogen fixation. These cells lack photosystem II, the complex that produces O₂ during oxygenic photosynthesis.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 7.8

17) Your microbiology professor is interested in isolating a particular endospore-forming bacterium from its natural environment and asks you to devise a protocol to do this. Explain how you could go about separating the endospores from the other microbes in the population.

Answer: The most direct method for isolating the endospores would be to expose the mixed population to high heat for a brief period to induce sporulation. This would also serve the purpose of killing any non-endospore forming bacteria.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 7.6

18) Various species of bacteria naturally produce antibiotics. What resistance mechanisms are utilized by bacteria to survive antibiotics produced by other microbes or from any antibiotics they may produce themselves?

Answer: Resistance mechanisms are genetically encoded and fall into four classes: (1) modification of the drug target; (2) enzymatic inactivation; (3) removal from the cell via efflux pumps, and (4) metabolic bypasses.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.10

19) Biofilm formation is known to increase the resistance of member bacteria to antibiotics. How does this happen?

Answer: There are several causes of increased antibiotic resistance in biofilms. These include decreased permeability of antibiotics due to the exo-polysaccharide matrix produced by the biofilm members. Efflux pumps also play a major role. In *E. coli*, genes encoding the AcrAB-To1C efflux pump are upregulated when cells enter biofilm growth. *Pseudomonas aeruginosa* encodes several multidrug efflux pumps that are more active when cells grow in an attached state.

Bloom's Taxonomy: 1-2: Remembering/Understanding

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